Explainable Survival Prediction and Biomarker Discovery in Hepatocellular Carcinoma

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***Abstract*—Accurate survival prediction in hepatocellular cancer remains challenging due to the complex interactions among clinical characteristics, tumor heterogeneity, treatment factors, and genomic alterations. Although machine learning has shown promising performance in improving prognostic prediction, the limited interpretability of many survival models reduces their clinical applicability and biological utility. This study proposes an explainability-driven survival learning framework that integrates clinical variables, tumor characteristics, somatic mutation profiles, and copy number alteration features from the HCC MSK 2024 cohort available through cBioPortal. The framework employs a leakage-aware pipeline with survival-driven univariate Cox feature selection, followed by matched Cox proportional hazards (CoxPH) and DeepSurv models in an effort to enable a fair comparison between statistical and neural survival learning. Model performance is assessed using Concordance Index (C-Index), integrated Brier score (IBS), time-dependent area under the curve (AUC), Kaplan–Meier survival analysis, and log-rank test. To improve model transparency, the proposed framework combines the intrinsic interpretability of CoxPH with three complementary explainable artificial intelligence techniques—DeepLiftShap, Integrated Gradients, and GradientSHAP—followed by consensus gene ranking to identify robust candidate prognostic biomarkers. Experimental results demonstrate that explainability-driven survival learning can improve the interpretability of prognostic models without sacrificing competitive predictive performance, and offer a practical framework for clinically meaningful survival prediction and biomarker discovery in hepatocellular cancer.**

***Index Terms*—Survival Analysis, Explainable Artificial Intelligence (XAI), Hepatocellular Cancer, DeepSurv, Cox Proportional Hazards, Cancer Genomics, Prognostic Modeling, Biomarker Discovery.**

I. Introduction

Primary liver cancer is still a major global health challenge. Hepatocellular carcinoma (HCC) accounts for nearly 75–85% of all primary liver cancers. Liver cancer is the sixth most commonly diagnosed cancer and the third leading cause of cancer-related death worldwide, indicating a poor prognosis and significant clinical burden according to the Global Cancer Observatory (GLOBOCAN) 2020 [1]. Despite advances in surgery, targeted therapy, and immunotherapy, survival outcomes still vary widely because of interconnected effects of demographic factors, liver function, tumour characteristics, treatment history, and underlying genomic alterations. Therefore, precise survival prediction is still a difficult but clinically important task.

Traditional prognostic evaluation is mainly based on clinical staging systems such as the Barcelona Clinic Liver Cancer (BCLC) system and the Tumor-Node-Metastasis (TNM) classification, and statistical survival models such as the Cox Proportional Hazards (CoxPH) model [2], [3]. These approaches provide clinically interpretable risk estimates; however, they rely on the assumptions of proportional hazards and linear covariate-survival outcome relationships, which restrict their ability to model complex nonlinear interactions in high-dimensional clinical and genomic data [4].

Deep survival learning models such as DeepSurv have been developed to capture nonlinear prognostic relationships by substituting the linear CoxPH risk function with deep neural networks, facilitated by recent advances in artificial intelligence [4]. Moreover, multimodal survival learning that leverages clinical, pathological and genomic information has demonstrated promising improvements in prediction performance [5], [6]. However, their implementation in clinical practice is still limited because their predictions are often not interpretable. In healthcare, reliable prognostic models should not only be highly accurate in prediction but also provide transparent explanations that allow clinicians to understand the factors influencing patient risk [7].

Explainable AI is an increasingly popular topic in survival analysis, but most existing work focuses on improving prediction performance and/or using a single post-hoc explanation method. Very few approaches offer an integrated framework that combines survival-aware feature selection, statistical survival modelling, deep survival learning, and multi-method explainability. Furthermore, few studies have explored whether multiple attribution methods consistently identify the same prognostic biomarkers, thus reducing confidence in the biological relevance of model explanations.

To address these obstacles, this work proposes an explainability-aware survival learning framework for hepatobiliary cancer using multimodal clinical and genomic data from the cBioPortal hepatocellular carcinoma cohort. The proposed framework combines three complementary feature-selection strategies with matched CoxPH and DeepSurv mod-

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els that allows a fair comparison of interpretable statistical modelling and nonlinear deep survival learning. We also integrate multiple explainability techniques, such as DeepLiftShap, Integrated Gradients, GradientSHAP, and consensus gene ranking, to detect robust and clinically meaningful prognostic biomarkers. The main contributions of this work can be summarised as follows:

- **A survival-driven feature engineering framework** based on Univariate Cox regression with hazard ratio estimation and Benjamini–Hochberg False Discovery Rate (FDR) correction is proposed for identifying statistically significant prognostic genomic biomarkers.
- **A fair comparative evaluation** of the classical CoxPH model and the DeepSurv model is performed using an identical set of survival-selected genomic features and multiple survival evaluation metrics, including C-index, IBS, time-dependent AUC, Kaplan–Meier analysis, and the log-rank test.
- **A comprehensive explainability framework** is developed by integrating DeepLiftShap, Integrated Gradients, and GradientSHAP to improve the interpretability of DeepSurv predictions.
- **A consensus gene ranking strategy** is introduced to identify robust prognostic biomarkers consistently supported across multiple explainability techniques.
- **The identified prognostic genes are interpreted** using CoxPH hazard ratios and validated against published HCC literature to improve the biological relevance and clinical interpretability of the proposed framework.

The remainder of this paper is organized as follows. Section II presents the related work. Section III describes the proposed methodology, including data preprocessing, feature selection, survival modeling, and explainability analysis. Section IV details the experimental setup and presents the evaluation results. Section V discusses the biological and clinical implications of the findings, and Section VI concludes the paper with future research directions.

II. Related Work

Recent development in machine learning and deep learning have resulted in a dramatic improvement in prediction for survival in hepatocellular carcinoma (HCC). Lee et al. developed a clinical decision support system based on machine learning for treatment recommendation and survival prediction. Wang et al. showed that deep survival models, such as DeepSurv, outperformed traditional statistical methods, such as CoxPH, on the SEER dataset [8], [9]. Similarly, Seven et al. and Zhao et al. investigated the advanced deep learning architectures for HCC prognosis and showed the improved predictive performance by feature selection and attention-based survival learning. However, these research have largely emphasized on the prediction accuracy and largely ignored the model interpretability and the discovery of biomarkers [10], [11]. Imaging based prediction of survival has also gained attention. Huang et al. proposed a whole-liver deep learning framework based on contrast-enhanced CT images for recurrence-free survival prediction, and showed better performance compared with traditional clinical models. These approaches, however, were mainly radiological-based and exhibited poor integration of genomic alterations or multimodal molecular biomarkers [12]. As artificial intelligence in oncology is quickly advancing, the necessity for explainable AI (XAI) is rising to enhance model interpretability. Ter-Minassian et al. proposed a survival-specific explainability framework based on MedianSHAP . Nalela et al. combined explainable machine learning with clinical and genomic data to predict cancer survivability [13], [14]. These studies enhanced interpretability of survival models, but were mostly limited to one attribution method and did not explore consistency of biomarkers across multiple explainability techniques. Additionally, recent work by Tan et al. showed that traditional machine learning methods still perform better than large language models for structured HCC survival prediction, emphasising the significance of task-specific survival modelling [15]. However, the existing works mainly focus on the predictive performance or explainability of a single method. Few works unify survival-aware feature selection, multimodal clinical and genomic data, matched CoxPH and DeepSurv modelling, and multi-method explainability in a single framework. Moreover, consensus-based biomarker discovery for hepatobiliary cancers remains underexplored. To fill these gaps, the proposed framework combines survival-aware feature selection with matched CoxPH and DeepSurv models, and then DeepLiftShap, Integrated Gradients, GradientSHAP, and consensus gene ranking to enable accurate survival prediction and biologically meaningful prognostic interpretation.

III. Methodology

A. Proposed Framework

In this study, we propose an explainability-driven survival learning framework for prognostic modelling of hepatocellular cancer using multimodal clinical and genomic data. The framework integrates clinical variables, tumour and sample features, somatic mutation profiles, and copy number alteration (CNA) features into a leakage-aware pipeline. The survival-selected genomic features are used to train both CoxPH and DeepSurv models under identical experimental settings, enabling a fair comparison between classical statistical and deep learning-based survival analysis. The trained DeepSurv model is then interpreted using multiple explainable artificial intelligence (XAI) techniques to identify clinically meaningful prognostic biomarkers, while CoxPH provides complementary statistical interpretation through hazard ratios and regression coefficients. The overall workflow of the proposed framework is shown in 1.

B. Dataset Description

The proposed framework was developed on the publicly available hcc_msk_2024 cohort cohort obtained from cBioPortal platform [7]. The dataset consists of multimodal clinical and genomic data including demographic, clinicopathological, treatment-related, somatic mutation and copy number

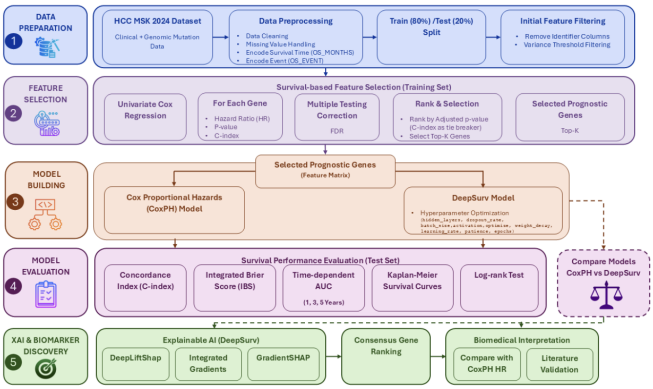


Fig. 1: Proposed methodology for survival prediction.

alteration (CNA) features. Overall survival was defined by OS_MONTHS and OS_EVENT. After excluding missing data, the final cohort of 1,357 patients with complete survival data was randomly divided into 1,085 training and 272 testing samples. After processing, the dataset contained 1,192 predictor variables.

C. Data Preprocessing

The dataset was divided into training (80%) and testing (20%) subsets using stratified sampling on the survival event to avoid information leakage. The preprocessing of data was done on the training set only and then applied to the testing set. The missing numerical values were imputed with the median. The categorical variables were imputed with the most frequent category and one-hot encoded . Then an initial feature filtering stage was performed by removing identifier related variables and applying variance threshold filtering to remove non-informative features before survival based feature selection. The processed dataset obtained was then used for further feature selection and survival modelling.

D. Feature Selection

A survival-aware feature selection using univariate Cox regression was performed on the training set only. Each predictor was fit separately into a CoxPH model and variables were ranked according to their statistical association with overall survival. Correction for multiple hypothesis testing was performed using False Discovery Rate (FDR). The final prognostic feature subset was chosen by ranking predictors according to the adjusted p-value, with the concordance index used as tie-breaker when necessary. The features with the highest scores were retained to construct the prognostic feature matrix for the subsequent survival modelling.

E. Model Building

The genomic features selected through the Univariate Cox feature engineering pipeline were used to train both the CoxPH and DeepSurv models. The CoxPH model served as an interpretable statistical baseline for survival prediction. It is one of the most widely adopted survival analysis methods in clinical research because of its ability to model censored survival data while providing regression coefficients and hazard ratios that quantify the effect of individual prognostic variables on patient survival. To improve model generalization and reduce overfitting, the CoxPH model incorporated L2 regularization, and the regularization parameter was optimized through cross-validation.

The DeepSurv model was employed to capture complex nonlinear relationships among the selected clinical and genomic features. Unlike CoxPH, DeepSurv replaces the linear risk function with a feedforward neural network while optimizing the Cox partial likelihood objective. Hyperparameter optimization was performed using cross-validation to determine the optimal network architecture and training configuration, including the number of hidden layers, neurons per layer, activation function, learning rate, optimizer, dropout rate, batch size, weight decay, training epochs, and early stopping strategy. The optimal hyperparameter configuration was subsequently retrained using the complete training dataset.

Training both models on the same survival-selected feature set ensures a fair comparison, allowing differences in predictive performance to be attributed to the underlying modelling approaches.

F. Model Evaluation

The trained survival models were evaluated on an independent test dataset using multiple complementary performance metrics. Predictive discrimination was assessed using the Concordance Index (C-index), while overall prediction accuracy across the follow-up period was evaluated using the Integrated Brier Score (IBS). The time-dependent Area Under the Receiver Operating Characteristic Curve (td-AUC) was computed at 1-, 3-, and 5-year survival horizons to assess the temporal discriminative ability of the models.

To evaluate clinical risk stratification, patients were categorized into high-risk and low-risk groups based on their predicted risk scores. Kaplan–Meier survival analysis was then performed to compare the survival distributions of the two groups, and the statistical significance of the observed differences was assessed using the log-rank test.

G. Explainable AI & Biomarker Discovery

The Univariate Cox feature engineering pipeline provides a biologically meaningful foundation for model interpretation by selecting statistically significant prognostic genes through survival-aware analysis. The intrinsic interpretability of the Cox Proportional Hazards (CoxPH) model was assessed using regression coefficients and hazard ratios, while the trained DeepSurv model was interpreted using three complementary explainable artificial intelligence (XAI) techniques: DeepLiftShap, Integrated Gradients, and GradientSHAP. Feature attribution scores from these methods were subsequently integrated through a consensus gene-ranking strategy to identify robust candidate prognostic biomarkers. Finally, the consensus-ranked genes were compared with the hazard ratios obtained from the CoxPH model and validated with published literature.

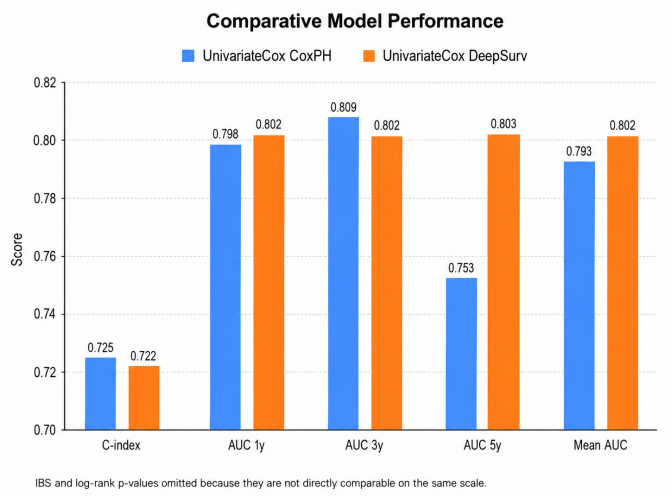


Fig. 2: Comparison of prediction scores (C-index, 1-year, 3-year, 5-year, and Mean AUC) for Univariate Cox feature selection pipeline combining with CoxPH and DeepSurv survival models

IV. Results

A. Comparative Performance Evaluation

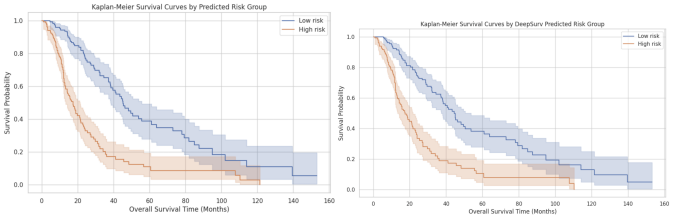
We evaluated the predictive performance of the proposed survival learning framework using the Concordance Index (C-index), Integrated Brier Score (IBS), time-dependent Area Under the Curve (td-AUC), and log-rank significance.

The proposed Univariate Cox feature engineering framework provides survival-aware feature selection, statistical interpretability through hazard ratios, and a biologically meaningful basis for explainable AI and biomarker discovery.

Figure 2 shows a comparative visualisation of the evaluation metrics across Univariate Cox feature slection pipeline combined with CoxPH and DeepSurv algorithems for survival analysis. The results indicate that both models provide competitive survival prediction when trained on the same survival-selected genomic features. While CoxPH offers slightly better global ranking performance and greater statistical interpretability through hazard ratios, DeepSurv demonstrates improved discrimination for long-term survival, highlighting the potential benefits of nonlinear deep survival learning when combined with survival-aware feature engineering.

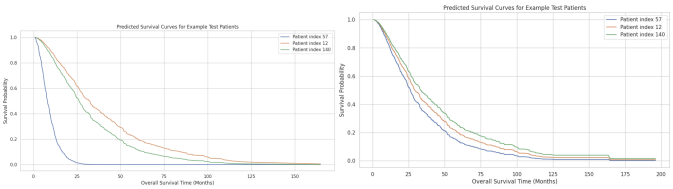
B. Kaplan–Meier Survival Analysis

Kaplan-Meier survival analysis was conducted to assess the clinical utility of the predicted risk scores. The survival curves generated by CoxPH and DeepSurv for the chosen explainability branch are shown in Figure 3. Risk stratification was effective as in both models, patients categorised as high-risk had a significantly more rapid fall in survival probability than those in the low-risk group. The survival curves separated early and remained separated throughout the follow-up period, showing that both models could discriminate patients with significantly different prognostic outcomes. The wider confidence



(a) (b)

Fig. 3: Comparison of Kaplan–Meier survival curves for low-risk and high-risk patient groups predicted by the DeepSurv model. (a) Kaplan–Meier survival curves stratified by the predicted risk groups, (b) Kaplan–Meier survival curves generated using the DeepSurv-predicted risk groups,



(a) (b)

Fig. 4: Predicted survival curves for three example test patients using (a) DeepSurv and (b) CoxPH. The curves show the estimated survival probability over time for each patient.

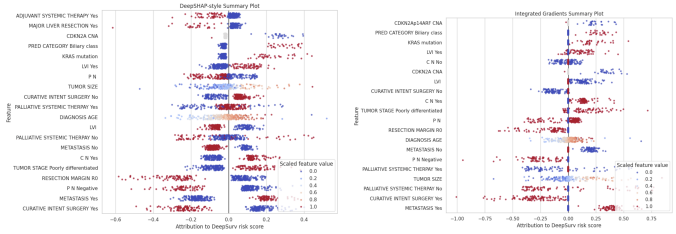
intervals at later time points are simply a consequence of the decreasing number of patients in follow-up, which is to be expected in survival studies. Statistically significant log-rank p-values also support the predicted risk groups.

C. Time-Dependent Survival Prediction

We also assess the individual survival prediction by visualising the predicted survival trajectories of representative test patients in Figures 4 for CoxPH and DeepSurv models, respectively. Both models gave patient-specific probabilities of survival which fell slowly over time with increasing estimated risk level. The DeepSurv model demonstrated improved separation of the predicted survival curves, especially for high-risk patients, suggesting stronger individualised risk differentiation. Conversely, CoxPH produced smoother survival trajectories with consistent ordering of patient prognosis. The results show that the proposed framework is able to provide personalised survival predictions, instead of uniform survival predictions for all patients.

D. Explainability Analysis

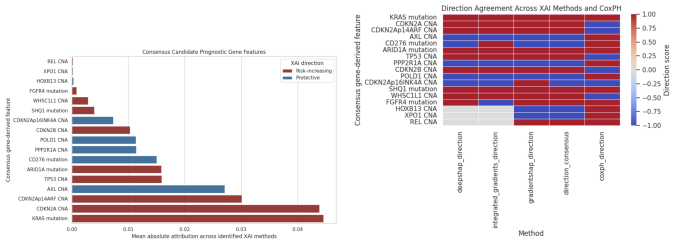
Three complementary attribution methods (DeepLiftShap, Integrated Gradients and GradientSHAP) were used to evaluate interpretability Figure 5. Despite the differences in the magnitude of the attribution, all three approaches were able to consistently identify multiple clinically relevant genomic features contributing to the predicted survival risk.



(a) (b)

(c)

Fig. 5: Feature importance plots generated using (a) GradientSHAP, (b) Integrated Gradients, and (c) DeepSHAP for explaining the DeepSurv risk prediction model.



(a) (b)

Fig. 6: Analysis of consensus gene-derived features identified by explainable artificial intelligence (XAI) models. (a) Mean absolute attribution across the evaluated XAI methods highlighting candidate prognostic features, categorized as either risk-increasing (red) or protective (blue). (b) Heatmap tracking direction agreement across individual XAI implementations (DeepSHAP, Integrated Gradients, GradientSHAP), their collective direction consensus, and comparison with a standard CoxPH baseline.

E. Consensus Gene Analysis

To increase robustness of biomarker identification, the feature attributions from the three explainability methods were combined with a consensus gene-ranking strategy. The consensus between these explainability techniques indicates that the identified feature attributions are robust and not dependent on a single explanation method. This multi-method approach provides increased confidence in interpreting DeepSurv predictions and a more reliable explanation of the learned prognostic patterns. The consensus ranking resulting from this is presented in Figure 6. The consensus on the direction of attribution between the XAI methods and CoxPH is shown in Figure 6b. Agreement in attribution direction between DeepSurv explanations and CoxPH hazard ratios increases confidence that the identified biomarkers represent biologically meaningful survival associations rather than model-specific artifacts.

Consensus analysis identified KRAS mutation, CDKN2A CNA and CDKN2A-p14ARF CNA consistently as the most influential risk-associated features across the evaluated attribution methods. Other genes with stable contributions to the model predictions included ARID1A, TP53 and AXL. Although some features had consistent attribution directions across DeepSurv explanation methods, there were differences between DeepSurv and CoxPH for a subset of genes, which suggests that nonlinear survival learning captures interactions that may not be represented by the linear assumptions of CoxPH.

To summarise, the explainability analysis shows that the proposed framework not only achieves accurate survival prediction but also provides biologically relevant interpretations through consensus feature attribution, leading to increased transparency and potential clinical usefulness of AI-assisted prognostic modeling.

V. Discusssion

The proposed framework shows that survival-aware feature selection with the UnivariateCox pipeline provides a good basis for explainable survival analysis in hepatocellular cancer. Unlike traditional survival approaches that primarily emphasise predictive accuracy, the framework aims to identify clinically meaningful prognostic factors through interpretable statistical modelling and consensus explainability, providing both survival-risk estimation and biologically relevant insights for future precision oncology applications. Although CoxPH is preferable when simplicity and direct hazard-ratio interpretation are prioritized, DeepSurv is justified when nonlinear relationships and interactions among clinical and genomic variables may influence survival risk, particularly for explainability-driven analysis of nonlinear feature contributions.

The consensus explainability analysis revealed 17 biomarkers (15 unique genomic loci) consistently identified by DeepLiftShap, Integrated Gradients, and GradientSHAP. Importantly, CDKN2A, CDKN2A-p14ARF and CDKN2A-p16INK4A are derived from a single genomic locus [16]. The agreement of three independent explainability methods indicates that the identified biomarkers represent robust survival-associated signals rather than artifacts of an individual attribution method.

Importantly, several consensus biomarkers identified by the proposed framework have already been validated in hepatocellular carcinoma. TP53 mutations are linked to poor overall survival [17], AXL overexpression is associated with tumour recurrence and decreased postoperative survival [18], and decreased ARID1A expression correlates with tumour metastasis and adverse prognosis [19]. Similarly, POLD1 and XPO1 have been reported as potential prognostic markers, with POLD1 associated with genomic instability and poor survival and

XPO1 with tumour progression [20], [21]. Agreement with published biological evidence supports the clinical relevance of the identified biomarkers and indicates that the framework captures meaningful molecular signatures rather than dataset-specific prediction patterns.

VI. Conclusion

We presented a leakage-aware, explainability-driven survival analysis framework for prognostic modelling of hepatocellular carcinoma by integrating clinical and genomic data. The proposed framework employs survival-aware Univariate Cox feature engineering to identify statistically significant prognostic genes, followed by a comparative evaluation of CoxPH and DeepSurv models using an identical set of survival-selected features. This design enables a fair assessment of the trade-off between predictive performance and model interpretability. Furthermore, the integration of CoxPH-based statistical interpretation, multiple DeepSurv explainable AI techniques, and consensus gene ranking facilitates the identification of clinically meaningful prognostic genes and robust candidate biomarkers associated with patient survival. The proposed framework demonstrates that survival-aware feature engineering combined with multi-method explainable AI provides a strong foundation for biologically interpretable survival prediction without compromising predictive performance. By integrating statistical survival modelling, deep survival learning, and complementary explainability techniques within a unified workflow, the framework supports robust prognostic biomarker discovery and has the potential to assist precision oncology and clinical decision support.

VII. Future Work

Future work will concentrate on validating the proposed framework on independent external cohorts and on reinforcing the biological interpretation of the identified consensus biomarkers. To explore the molecular mechanisms of the identified prognostic genes, we will perform extensive functional analyses, including STRING protein–protein interaction networks, Gene Ontology (GO) enrichment, KEGG, and Reactome pathway analyses. Furthermore, the framework will be expanded towards an interactive clinical decision-support platform that can offer personalised survival prediction with feature-level explanations, thus enabling its potential translation into real-world clinical practice.

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